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MACSE504 - Machine Learning

Course Based Design Project

Date: 30.10.2026 **Slot:** E1+TE1

Your uploaded file must contain:

→ Registration Number, Name, and Question text.
→ Provide a brief description and pipeline/architecture of the project.
→ Include source code paired with its corresponding output.
→ Provide an inference on model parameters, evaluation of performance metrics, and the final solution to the problem statement.
Note: Refer to the table below to find your assigned question.

Questions:

1. **Healthcare Analytics (Malignant vs. Benign Tumour Classification)**
Medical practitioners require not only high accuracy in classifying tumours but also transparent decision boundaries. Ingest cytological feature measurements, construct competing Linear SVM, RBF-Kernel SVM, and Decision Tree architectures, and critique their structural complexity against feature importance rankings.

***Dataset Link*:** UCI Breast Cancer Wisconsin (Diagnostic) Dataset

Requirements:
i. Data profiling, scaling, handling multicollinearity, and stratifying train/test splits.
ii. Implementation of baseline Decision Tree, Linear SVM, and Non-Linear RBF SVM.
iii. Hyperparameter search (tree depth vs. SVM soft-margin cost C and kernel parameter γ. Extract feature importances.
iv. Comparative analysis using ROC-AUC charts, Confusion Matrices, and a runtime breakdown.

2. **Regularized Regressors for Volatile Asset Pricing**
Predicting asset pricing profiles requires structural defence against highly correlated variables and high-dimensional noise. Apply explicit polynomial feature extensions to capture non-linear properties and use structural regularization paths (Lasso/Ridge penalty sweeps) to systematically drive redundant coefficients down to zero.

Dataset Link: Ames Housing Dataset via Kaggle

Requirements:
i. Log-transforming target variables, handling missing features, and setting up one-hot encoders.
ii. Generating a 2nd-degree polynomial matrix expansion and fitting an unpenalized OLS Multi-Linear model.
iii. Implementing Lasso and Ridge. Grid-searching the optimal regularizer strength alpha (α).
iv. Performance benchmarking on test sets using absolute errors (MAE) and root residual errors (RMSE).

3. **High-Dimensional Customer Micro-Segmentation**
   Developing targeted banking campaigns requires groupings across diverse, correlated financial habits. Reduce transaction dimensions using variance-retaining PCA projections before applying iterative cluster allocation that sweeps to locate optimal behavioural segments.

   ***Dataset Link*:** Kaggle Credit Card Dataset for Clustering

   Requirements:
   i. Robust scaling and executing PCA to retain at least 85% of global dataset variance.
   ii. Executing K-Means for K in [2, 12]. Generating Within-Cluster Sum of Squares (WCSS) profiles.
   iii. Mathematically validating the elbow inflection point against Silhouette Coefficient profiles.
   iv. Plotting clustered customer spaces onto top principal axes and profiling segment centres.

4. **Genomic Taxonomy Mapping via Dissimilarity Matrices**
   Uncovering hierarchical structures in biometric features requires evaluating various link metrics. Compute complete pairwise similarity distance maps, generate divergent linkage tree networks, and use non-linear projections to see how well these structures match real categorical groups.

   Dataset Link: UCI Iris Dataset (or TCGA Pancancer Gene Expression)

   Requirements:
   i. Generating Agglomerative trees across diverse linkages (Single, Complete, Average, and Ward).
   ii. Visualizing linkage trees and identifying truncation lines using Cophenetic Correlation coefficients.
   iii. Running a t-SNE manifold learning projection to map high-dimensional layouts to 2D.
   iv. Cross-checking hierarchical groupings against t-SNE visual clusters to evaluate spatial consistency.

5. **Transactional Pattern Mining in Retail Analytics**
   Optimizing product cross-selling strategies requires analyzing large, sparse transaction datasets. Map invoice records into boolean arrays and evaluate structural speed and rule quality differences between index sweeps and recursive prefix trees.

   Dataset Link: UCI Online Retail Dataset

   Requirements:
   i. Cleaning invoice streams, dropping cancellations, and formatting records into sparse sparse matrices.
   ii. Implementing the Apriori pipeline alongside an optimized FP-Growth tree layout.
   iii. Extracting rules based on Support, Confidence, and Lift thresholds.
   iv. Profiling computation speeds and memory usage across varying minimal support thresholds.

6. **Semi-Supervised Classification of Partially Labeled Data**
   In real-world settings, labeling data completely can be prohibitively expensive. Convert raw text streams into TF-IDF vector matrices, simulate a limited annotation budget by masking 90% of the training labels, and evaluate how graph proximity methods compare to pseudo-labeling pipelines.

   Dataset Link: UCI SMS Spam Collection Dataset

Requirements:

i. Tokenization, removing stop words, TF-IDF vectorization, and masking training labels.
ii. Building a Self-Training framework using a base Logistic Regression classifier.
iii. Implementing RBF-Kernel graph propagation and Label Spreading architectures.
iv. Evaluating classification metrics (Precision, Recall, F1) against a fully supervised baseline.

7. **Complexity Reduction Over Structural Coordinate** Spaces
Deploying image recognition models on edge hardware requires balancing computational throughput with structural accuracy. Students will implement singular value decompositions to compress pixel spaces and analyze the trade-offs between component counts, model memory footprints, and classification accuracy.

Dataset Link: MNIST Digit Dataset via Kaggle

Requirements:

i. Visualizing eigenvectors (eigen-digits) and plotting cumulative explained variance ratios.

ii. Constructing a scikit-learn pipeline coupling PCA dimensionality reduction with a Logistic Regression classifier.

iii. Running grid-searches across component counts k in [5, 50, 100, 200] using stratified cross-validation.

iv. Generating curves mapping feature reduction levels against classification latency and test errors.

Question Allotment:

REGISTER NO	NAME	Q.No
25PHD2036	BODATHOTI ESWARAIAH	1
26MML0001	SHIVANI RAJPUT	2
26MML0002	ANUBHA SHUKLA	3
26MML0003	PRAKHAR AGRAWAL	4
26MML0004	RITESH RAJ	5
26MML0005	SANJANA G S	6
26MML0006	VANA SHIVARAMAKRISHNA KUMAR	7
26MML0007	SARANYA DALAL	1
26MML0008	LIKHITH U	2
26MML0009	DIVYA VIJAYKUMAR PUJARI	3
26MML0010	GEDELA BALAKRISHNA	4
26MML0011	CHRIS JOY	5
26MML0012	CHINRAJ A	6
26MML0013	DEBJIT DEY	7
26MML0014	URBI CHAKRABORTY	1
26MML0015	K V KARTIK	2
26MML0016	ASWIN DINESH NAIR	3
26MML0017	GUDIWADA NAGA LAKSHMI NAVEEN	4
26MML0018	BHUVANESHWAR S	5
26MML0019	ABHISHEK P	6

26MML0020	ROHAN YADAV	7
26MML0021	SATYAYAN SINHA ROY	1
26MML0022	SHREYAN PANDA	2
26MML0023	JESWIN J	3
26MML0024	PATEL CHELSI JAYENDRAKUMAR	4
26MML0025	JITESH NITEEN BOBADE	5
26MML0026	DADHANIA PRIT SHAILESHBHAI	6
26MML0028	SHANJEEV KUMAR G	7
26MML0030	Janapati Venkata Sai Chandra Kousthubh	1
26MML0031	MAKADA HASNAIN HUSAINBHAI	2
26MML0032	ADDWIN VIBRO D	3
26MML0033	SHREYAS N GATTI	4
26MML0034	VAISHNAVI AJEET JADHAV	5
26MML0035	RIVIN JOSE MELVIN	6
26MML0036	AVALOORU SREEJA	7
26MML0037	BEZWADA RAJ DEEP	1
26MML0038	ABHISHEK P	2
26MML0039	JOSHI VIRA BHAVESH	3
26MML0041	GEDDAM DEEPIKA	4
26MML0042	JOSPAR MILLIAN J	5
26MML0043	AYUSH SHARMA	6
26MML0044	PATIL RAMRAJE NAGESH	7
26MML0045	NIMKAR VEDANT GAJANAN	1
26MML0046	ARYAN KISHOR SAO	2
26MML0047	VINAYAK C	3
26MML0049	SAHIL MALL	4
26MML0050	IYSWARYA P	5
26MML0051	BAROTE SUJEET JAGANNATH	6
26MML0052	CHILLA KALYAN SAI KUMAR REDDY	7
26MML0053	EDVIN SAJI	1
26MML0054	MUDAVATH NARASIMHA NAIK	2
26MML0055	MAHAJAN DURVESH CHANDRAKANT POOJA	3
26MML0056	VALIVETI SAI SRINIVAS	4
26MML0057	CHANIYARA TARANGKUMAR PARESHBHAI	5
26MML0058	PRIYANKA R	6
26MML0059	K KARISHMA	7
26MML0060	DEVI PRIYA K	1
26MML0061	ARJUN AJIT	2
26MML0062	MADHUVANTH H	3
26MML0063	PRIYADHARSHINI K	4
26MML0064	R G RITHICK RAGHAV	5
26MML0065	PRASANNA KUMAR M	6
26MML0066	LATHA PRABHA S	7
26MML0067	SINDHIYA G	1
26MML0068	KISHORE V	2